

CTL Key Encapsulation Algorithm x86 Architecture Implementation Self-Evaluation Test Report

Algorithm Name: CTL

Algorithm Category:KEM

Sub-functions: Key Generation / Encapsulation / Decapsulation

Implementation Version: Optimized Implementation Version

Security Levels:128bit/256bit/512bit

Contents

1. Overview.....	3
2. Test Environment	3
2.1. Hardware and Software Configuration	3
2.2. Test Tools.....	3
3. Optimized Implementation Test	4
3.1. Functional Test.....	4
3.2. Performance and Resource Consumption Test	65
3.3. Transmission and Storage Overhead Test	67

1. Overview

This report conducts self-evaluation tests on the optimized implementation version of the CTL key encapsulation algorithm under the x86 architecture. The test scope covers functional test, performance and resource consumption test, as well as transmission and storage overhead test.

The functional test verifies the integrity of core components of CTL algorithm including key generation, encapsulation and decapsulation. It also validates the consistency between the shared secret generated by the encapsulation side and the shared secret recovered by the decapsulation side. Fixed random seeds or fixed test vectors are adopted during testing to ensure reproducible test results.

The performance and resource consumption test counts the average time consumption, average CPU cycles and throughput of three core operations: key generation, encapsulation and decapsulation. It also records the static memory footprint after algorithm loading and peak runtime memory usage. Each test shall be executed no less than 100 times; the actual number of test runs shall be filled in according to test commands.

The transmission and storage overhead test counts the data sizes involved in the CTL key encapsulation algorithm, including public key, private key, ciphertext and shared secret value.

2. Test Environment

Test Period:2026/6/24 - 2026/6/25

2.1. Hardware and Software Configuration

Serial No.	Name	Specifications & Identifiers	Purpose	Number
1	CTL Optimized Implementation Source Code	Version: 1.0; Parameter Sets: 257-512 / 769-1024	Program under test	1
2	Test Terminal PC	CPU: Intel® Core™ i9-14900HX; Main Frequency: 2.2GHz; Memory: 32 GB; OS: Windows11	Compile & execute test programs	1
3	MinGW Compilation Environment	GCC Version: 15.2.0; Target Architecture: x86-64	Compile C test programs	1

2.2. Test Tools

Serial No.	Tool Name	Tool Type
1	MinGW / MinGW-w64	C Language Compilation Tool
2	benchmark_selfeval_ctl.c	Performance, resource consumption, transmission & overhead test program
3	test_kem_correctness_ctl.c	Functional correctness test program

3. Optimized Implementation Test

This chapter covers tests for the optimized implementation of CTL key encapsulation algorithm.

Compilation commands used for optimized implementation:

```
gcc -O3 -march=x86-64 -mavx2 -mtune=native -flto -fomit-frame-pointer -std=c99 -Wall -Wextra -Wpedantic  
test_kem_correctness_ctl.c <CTL_SOURCE_FILES> -o test_kem_opt.exe
```

```
gcc -O3 -march=x86-64 -mavx2 -mtune=native -flto -fomit-frame-pointer -std=c99 -Wall -Wextra -Wpedantic  
benchmark_selfeval_ctl.c <CTL_SOURCE_FILES> -o benchmark_opt.exe -lpsapi
```

3.1. Functional Test

ID	OPT-1
Test Name	Functional Correctness Test for CTL-257-512 Optimized Implementation
Test Objective	Verify that the CTL-257-512 optimized implementation supports key generation, encapsulation, and decapsulation, and verify whether the original shared key is consistent with the decrypted shared key.
Preconditions	1. CTL optimized implementation source code has been deployed on the test terminal PC; 2. test_kem_correctness_ctl.c can normally invoke CTL's key generation, encapsulation and decapsulation interfaces; 3. Test program can compile and run normally.
Test Steps	1. Enter the directory of CTL-257-512 optimized implementation source code; 2. Execute make all to generate test_kem_correctness_ctl.exe; 3. Run test_kem_correctness_ctl.exe; 4. Compare the shared secret from encapsulation (ss_enc) and decapsulation (ss_dec) to check full consistency; 5. The algorithm will be executed three times in a row and output results to file ctl_kem_kat_vectors.txt.
Expected Results	1. test_kem_correctness_ctl.exe runs without error; 2. Terminal outputs "All CTL KEM correctness tests passed."; 3. Test vector file ctl_kem_kat_vectors is generated successfully; 4. ss_enc and ss_dec values in test vectors are identical.
Actual Results	The program runs normally and prints "All CTL KEM correctness tests passed." All three groups of ss_enc and ss_dec in ctl_kem_kat_vectors.txt are equal.

algorithm = CTL-257-512

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16

pk =

99B4BD408EE8A95011ACEA41FEE592AF5879CD3D3EAD420122FB50AC7B94D5F15091E56DA6D
57C7A10E2766DF8144E94A2D67DE507A6139B0DA5DC0B72A106A1FA49413C8612D88938AC9316
DC8A0EFB45464AD04FDA893F76B94333FDA9976DFBFD39CDC7CDB2BC071FD0F98D5551D5D35
A4CF0EFF3C04BF4EAA49013E3A3586A562C012D6CEBB7DCB48F1B310F471365D884FD22C9217F
36615FA02CD2082B0C33D7CA23014D918331530BE5E0D00598F7AF0B93E55A9151146B3D80889682
1903579022FF90557DBA44F95E401DD029E325F672D50CC90F0FEE91799F3F49E44FCEC1330B8EFC
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DA9B9202F4E3E791EA7DC7D953AF696588FE47907C8A1A908F5D2EAE969A9421029A9F7A664DB
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F9024946CB7E7D2ED3C6B7CFBF164A0D36E1521DD2EE461B9DE28AF9019A85D33DE8C72BC75E
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00920AD9FE9510F147A389E5BCD6BA907

sk =

19AE5D933F84538C0270359F7DED4695A38B68A9E050084EFBA8BDCDD685B942EE3FE05922EE90
1FE0E9E956242B81B7F61C15C1111D711EAEB4076508CEC087DD0000F011FF00F0011010000F00001
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0F101F00000001010F00011F00000F010101110F00F0000F000000110000000000000F0F0100FF0000001
F00F000000001001000100110F1000F00FF0F1F0F00F0010000010FF00FF000F0000F00F001F0000FF001
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B4BD408EE8A95011ACEA41FEE592AF5879CD3D3EADE420122FB50AC7B94D5F15091E56DA6D57
C7A10E2766DF8144E94A2D67DE507A6139B0DA5DC0B72A106A1FA49413C8612D88938AC9316DC
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CF0EFF3C04BF4EAA49013E3A3586A562C012D6CEBB7DCB48F1B310F471365D884FD22C9217F366
15FA02CD2082B0C33D7CA23014D918331530BE5E0D00598F7AF0B93E55A9151146B3D80889682190
3579022FF90557DBA44F95E401DD029E325F672D50CC90F0FEE91799F3F49E44FCEC1330B8EFCFA
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F00D6BABF8F807BFAF6062E8CFA2A79872931B5BE46164F97E0DAD625C100521488E6331E93F902
4946CB7E7D2ED3C6B7CFBF164A0D36E1521DD2EE461B9DE28AF9019A85D33DE8C72BC75E97E4
7E97DC02A7F1D94B3C3D54B994B59D2F48C30B6DDCFF0253E781E573CD09BFC1D20076C4587A4
F85009894AD762C19D7EE5FD0356B546F654705DEAB6426B0916A05057E176F2A00C5B2D34900920
AD9FE9510F147A389E5BCD6BA907

ct =

D98D3DAAFB6D25886B47F191AA1E96323CB0B48B185E1B7F0DAB551ADE7C6A0B7DF36741428C
BC7DE0FC30FEF88B3F62F4C11C82E28A7BA66E94BC4B099E72015F2CE8A95EC2A54371FA6535F1
696A7D385D9373B2705133E05CE35E3D8D350E5B06CC20CC451368F31D1C6E8D29B7F63C7F6CCA
2A19397E594B6FC51388B3A575C29A25BEEE271ADCE66477ABC252B7DDB9ADC3151663D8499EB
3FE32FBBE8EBB722E47DC16F542D139B939B635C1A77B9245E84EB4FA7DE57494A9EF6674707763
4CBFE531C6D231C23A292E39B4A3BCF5DDEAF27463F2AA1BAC9B643A840E93FCC54F4323F33E
B780A8AB5309112A27C953241AA31A78325D41204F0AEA2F7B8FB2C2EA3111831DF8052347E3C9
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9D59445A4CFE6F739D72B8AE7851473BB20B7DE12E167D0F9EC51FEF9F9467E21C5B0FFA09EF88
8F319CC8311745E685099A7CEC230B73F937E496B2D60B39F7FA2B419678B70A02114A91843DE800
F7AA58F6FFFA992AB3F847CC0A1E4B54DBCA405E65BBF0D577D01AD73F84D3E801A99157AC2A
3A3AE1E78A39284A9A0736242014B56AA4147D930F8C2826D080B9CE114A873906B90B1AE9E9876
E054B8D0C284

ss_enc = 460FE91C8D752E7BA74838C7B2B891DD

ss_dec = 460FE91C8D752E7BA74838C7B2B891DD

result = PASS

case = 2

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16

pk =

991581121AADB24C53C4E3C3ED26E30087DDA5E64C6C24CE94963F7C4BE3EBEA5C89F63B141F3
B9B1D37FB499539AD64C53B5502F1680A15631F6C98F730431373C9A54EFBA63DAB090E9725C196
1EE1CD771F4FB93A9E5786D8F1FA74BF773C4C4E48384E6CAE5392417570B9B85E49E1800BC4469
7A73EEF59C550B34D8509FD3E3B878A74A3BEA2E681C900DB223F2BDE9CF48A56A945729486E7E
A53567320E28719AB60AB6DC218C85822BD0E5814AB3828656A180B93F22B6BEE3851234EE91440
7E4B59A6E7D59A604C8CACD5233B5EE7D5DEBCAF0FB8949C917DA888C0A366767058E64022C0F
6DBE36AA3DE5E0EC9C9891C11E86319FB71090EF54E588B3A30C14626FAB7888FD04440C94A178
B7FA5590D86C40CF4F9A8FC5D97E7BEAF4898385C05FE44C0560D7137DADEAE8BEA2A003F406
AE1B7E66F847080A86F56ECE1142826DBCC33D392F46495CB9B6DA18C5A16E38DE256C3CA1F20

AD186A4207A91B5EC6BBC546C963B9D645CA14C5A8255935277F87FC4993F6AEACCEE1550915C
4328A34F2D50E6A758035CC20718A915A679B31460BA273B3F64B19DBC8372A0999949AD412C290
E3205752F8AC15AAECF02D06F776FDD508418B8DA406229E53301F1170D1D026F8562B13E3B6C10
CDDBC2266F66AF635742D3E3568ABF55F0B437B732B81FE89A9D4A126DEC5EA5CEEB266EF9DA
60851CCBD3E522E3429600E8D552AD2

sk =

19C10DA35D778BACBD25A9A47315A4F312EFD4EC044C315DC4CA00F59E297A9F8D1A79ACA146
51B4FC7482B26D2F7125FCBEED20025EE7947A5AE5FE04F9B6681EFFFF000F0F0000000001F1FF1
00F00F0000F100000000F00000010010000000000F02F0F0F100100000FF0000F00F1000F1000F0100F
0010F0000000100000000000F001010001F0F0100F000F01E111000010F00100000F000F010FF1000000F
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46FC2E7C201F3CEBCFBCEFD3084FC1002088F02FFFF7FFC207BF7BFBE044F3ADC213C03DFC5
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52F8AC15AAECF02D06F776FDD508418B8DA406229E53301F1170D1D026F8562B13E3B6C10CDDDB
C2266F66AF635742D3E3568ABF55F0B437B732B81FE89A9D4A126DEC5EA5CEEB266EF9DA60851
CCBD3E522E3429600E8D552AD2

ct =

D95C4D11C8C700528D3B2F5F2F30BA1ED3D4476B23F7B92E711B412B5C6BB72EC5FE7578B9B891
236EAE25F198AD9793241FF28DEA7C2A0DE09537CC4C23B82E5E6E98818970CD76606F8E673347
1F26A476AD257A8C7F5D54A733F1EA399A25AAEDA311889C11E23C200418DD74EFAD09624A98B
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A6CB362094DCDF00B4A1A6A8BF3ACECF8D9579AE56BF913E41570A9A585C7737B42E383D225C
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44481B36874A523FE49982F5AC86DEBB091F88DB27D348B2E24980E6C399991D4DB683AAACDEE
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5D0001294DC01EE082E985488602A77E6CE77D6398A0610954D766206FF25E75BDB502CAA6C027C

84AD5A06317A13EC69AF9D92534A2CB42D06C65B70726876EB50FC41322FFBBDDBCA706004B2D8
DA89DB7E9D1D7431B97918445CFB3194DE3B9AB6E22657CEFA61B3702446613FC2E0600FA319C9
6CB3DBAC268

ss_enc = C5FD9F8003010BDA9A156C81FFD8B31D

ss_dec = C5FD9F8003010BDA9A156C81FFD8B31D

result = PASS

case = 3

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16

pk =

998B97C194C0DD837E7803595B40DDC4A6E11507EEF4270CBE8C74BD12509A80E06A0D49D50B6
B83A05093BB09D63F2D1B1FE7E45FC0AF742D03BA43D094C18B1A60E65C5C3698E5C735BD74905
F6BC51E078A16D9C3A45D42818B58777FCCCF083A0236BD9E6627D33BDE2AA0AADE4B16C42B
AF89EF0A7ADBFF99C8269DF716D77328CF0C5BF54CE6A69F9762D763C40DBE52E0C015EBDA70F
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sk =

19DB89239991D32C3EB3108A23D1421031A8D9487C3780776C30D933302760C3740655D84B01E4B3
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BDFC103DF4303C1C0043F80FBCFFF0B8002204F7E04917F04007E0F8F81146EC1EFBF43084FFFF7F
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42FBDF7F0FC17FFC0FBDF3D0C313EF3FF01FC6E41F810C517E17FFFAF0803CFFA08A14103C0BFC
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3DB5D4E2E1792D3B4684B14000B0F208396395890047EA133301FEA5B121E94918D35FFB61F297198
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DC5C074EB5C71B1799DE2462046E0AFB0718A0BA512FC8FA38E6F0533D333B038DB22683BF1160
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81D14665402DBF8204F73603D4D5E375B0058F06C7D0CB10A39365B79658F760965633DC202B1119
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968A77295EBEF47075BDD336836DE4739F194B782091B227EEA110821ED87C731E0294AE18E7DCC
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93147017A80066BB96FF3BF454A40C888588FE873AE8570A11FF7071A70992C007036414C094B0DF6
1E6044E1447668A8074A2

ct =

D93290FD94A2C273DB3BD306FEB89A4D160733DF7E87361E96248F182A67CDC938FE415CBE4340
31D288090BDFD18F12F341862F0FD3C4F19AD8880555BD1A750D46D6FEE6FA81846BE3095768FE4
11A42F53A19FBEC2E5E38466678E43A20C2A5A3540007D34FB273AF66CA99F6AB26528C624DC85
E590AB276ECE6774CA65D9FCF8B196ECCD8454BF3D771999176FDB7659ECC5180A2D8CD9569B7
8A626E331B1046817D639D17C7D6CF45D02271ED0999DCB775BCEBE53D0EB4943271876A8904EA
67CD03FBFFFEFFCCC9C7C076BD42B150274D131D2722DD668B33934153C6248AAFC12D50A3F2F0
6FE104B2BFB2E3B171C7109FE7245A34E73D301BC1F6A07E129A109FD182681A5780E7C8A3AB6D
039ECFE8753AF3FC83EE0C6D52E79BD47B6F1522384B42E49ADA156F2EBC6762CB3B32201C79E4
9470D978268353A553162CA343B3FD01BF33E1EBC7141DCA9F19E2DD103220921CC8EBD905E58C
804C9F632DD3560BF78A21513DE403BDEDF4AA95EFE492C96AB37B739263ACE66BF0383915A036
D9F97E91D1F285C87688B0B1E793B8EC478517DFE074B3D456896C53392A846259B88C45BF71AF8
1200DA5853B796F9117AD03450DF117073203B1EBD19F7B67207808F6F71CCA8ABC211921C98F8F
96C6D1B29

ss_enc = 28A57455611066AE6E52E5E7A2604445

ss_dec = 28A57455611066AE6E52E5E7A2604445

result = PASS

ID	OPT-2
Test Name	Functional Correctness Test for CTL-769-1024 Optimized Implementation
Test Purpose	Test whether the optimized implementation version of CTL-769-1024 supports the functions of key generation, encapsulation and decryption, and verify whether the original shared key is consistent with the decrypted shared key.
Preconditions	1. CTL optimized implementation source code has been deployed on the test terminal PC;

	2. test_kem_correctness_ctl.c can normally invoke CTL's key generation, encapsulation and decapsulation interfaces; 3. Test program can compile and run normally.
Test Steps	1. Enter the directory of CTL-769-1024 optimized implementation source code; 2. Execute make all to generate test_kem_correctness_ctl.exe; 3. Run test_kem_correctness_ctl.exe; 4. Compare the shared secret from encapsulation (ss_enc) and decapsulation (ss_dec) to check full consistency; 5. The algorithm will be executed three times in a row and output results to file ctl_kem_kat_vectors.txt.
Expected Results	1. test_kem_correctness_ctl.exe runs without error; 2. Terminal outputs "All CTL KEM correctness tests passed."; 3. Test vector file ctl_kem_kat_vectors is generated successfully; 4. ss_enc and ss_dec values in test vectors are identical.
Actual Test Results	The program runs normally and prints "All CTL KEM correctness tests passed." All three groups of ss_enc and ss_dec in ctl_kem_kat_vectors.txt are equal.

algorithm = CTL-769-1024

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

pk =

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ss_dec = 1AEFFE6D4470D34D66A5A7D3AB59573E5B6B9176ACD1A34115A8194B3A8998BA

result = PASS

case = 2

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

pk =

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sk =

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result = PASS

case = 3

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

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ct =

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ss_enc = B11EF9E97DDC18BE412371D4279CB7626625E9B8459D5E25B0D625C289DDDAAC

ss_dec = B11EF9E97DDC18BE412371D4279CB7626625E9B8459D5E25B0D625C289DDDAAC

result = PASS

ID	OPT-3
Test Name	Functional Correctness Test for CTL-3329-2048 Optimized Implementation
Test Purpose	Test whether the optimized implementation version of CTL-3329-2048 supports the functions of key generation, encapsulation and decryption, and verify whether the original shared key is consistent with the decrypted shared key.
Preconditions	1. CTL optimized implementation source code has been deployed on the test terminal PC; 2. test_kem_correctness_ctl.c can normally invoke CTL's key generation, encapsulation and decapsulation interfaces; 3. Test program can compile and run normally.
Test Steps	1. Enter the directory of CTL-3329-2048 optimized implementation source code; 2. Execute make all to generate test_kem_correctness_ctl.exe; 3. Run test_kem_correctness_ctl.exe; 4. Compare the shared secret from encapsulation (ss_enc) and decapsulation (ss_dec) to check full consistency; 5. The algorithm will be executed three times in a row and output results to file ctl_kem_kat_vectors.txt.
Expected Results	1. test_kem_correctness_ctl.exe runs without error; 2. Terminal outputs "All CTL KEM correctness tests passed."; 3. Test vector file ctl_kem_kat_vectors is generated successfully; 4. ss_enc and ss_dec values in test vectors are identical.

Actual Test Results	The program runs normally and prints "All CTL KEM correctness tests passed." All three groups of ss_enc and ss_dec in ctl_kem_kat_vectors.txt are equal.
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algorithm = CTL-3329-2048

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 3009

sk_len = 15617

ct_len = 2353

ss_len = 48

pk =

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result = PASS

3.2. Performance and Resource Consumption Test

Test ID	OPT-4
Test Name	CTL-257-512 Optimized Implementation Performance and Resource Consumption Test
Test Objective	Test the average CPU cycles, throughput, static memory usage, and peak memory usage of the three core operations, key generation, encapsulation, and decapsulation, in the CTL-257-512 optimized implementation.
Preconditions	1. The CTL optimized implementation program code runs normally; 2. benchmark_selfeval_ctl.c can normally call the CTL algorithm interfaces; 3. The test program can be compiled and run normally.
Test Steps	1. Enter the CTL-257-512 optimized implementation program code directory; 2. Enter make all to generate benchmark_selfeval_ctl.exe; 3. Run benchmark_selfeval_ctl.exe to collect statistics on the total time, average time, average CPU cycles, and throughput for key generation, encapsulation, and decapsulation; 4. The program records static/baseline memory usage before the algorithm runs, records peak memory usage during repeated execution, and finally generates the ctl_kem_benchmark_raw.csv and ctl_kem_benchmark_summary.csv result files.
Expected Results	The program can run normally and output the algorithm performance test results, including the average CPU cycles, throughput, static/baseline memory usage, and peak memory usage for key generation, encapsulation, and decapsulation operations.
Actual Results	The program can run normally and outputs performance test results. The program collects statistics on the average CPU cycles, throughput, static/baseline memory usage, and peak memory usage of key generation, encapsulation, and decapsulation operations. The number of test runs is 100. Key generation: 7358433 cycles/op average, 328 ops/s throughput; Encapsulation: 19383 cycles/op average, 125407 ops/s throughput; Decapsulation 88852 cycles/op average, 27250 ops/s throughput. Static/baseline working set memory average: 5559042 Bytes; Peak memory average: 5562368.00 Byte;

Test ID	OPT-5
Test Name	CTL-769-1024 Optimized Implementation Performance and Resource Consumption Test
Test Objective	Test the average CPU cycles, throughput, static memory usage, and peak memory usage of the three core operations, key generation, encapsulation, and decapsulation, in the CTL-769-1024 optimized implementation.
Preconditions	1. The CTL optimized implementation program code runs normally; 2. benchmark_selfeval_ctl.c can normally call the CTL algorithm interfaces; 3. The test program can be compiled and run normally.
Test Steps	1. Enter the CTL-769-1024 optimized implementation program code directory; 2. Enter make all to generate benchmark_selfeval_ctl.exe; 3. Run benchmark_selfeval_ctl.exe to collect statistics on the total time, average time, average CPU cycles, and throughput for key generation, encapsulation, and decapsulation; 4. The program records static/baseline memory usage before the algorithm runs, records peak memory usage during repeated execution, and finally generates the ctl_kem_benchmark_raw.csv and ctl_kem_benchmark_summary.csv result files.
Expected Results	The program can run normally and output the algorithm performance test results, including the average CPU cycles, throughput, static/baseline memory usage, and peak memory usage for key generation, encapsulation, and decapsulation operations.
Actual Results	The program can run normally and outputs performance test results. The program collects statistics on the average CPU cycles, throughput, static/baseline memory usage, and peak memory usage of key generation, encapsulation, and decapsulation operations. The number of test runs is 100. Key generation: 28096354 cycles/op average, 86 ops/s throughput; Encapsulation: 28599 cycles/op average, 84875 ops/s throughput; Decapsulation 169168 cycles/op average, 14306 ops/s throughput. Static/baseline working set memory average: 5792311 Bytes; Peak memory average: 5801041.92 Byte;

Test ID	OPT-6
Test Name	CTL-3329-2048 Optimized Implementation Performance and Resource Consumption Test
Test Objective	Test the average CPU cycles, throughput, static memory usage, and peak memory usage of the three core operations, key generation, encapsulation, and decapsulation, in the CTL-3329-2048 optimized implementation.
Preconditions	1. The CTL optimized implementation program code runs normally; 2. benchmark_selfeval_ctl.c can normally call the CTL algorithm interfaces; 3. The test program can be compiled and run normally.
Test Steps	1. Enter the CTL-3329-2048 optimized implementation program code directory; 2. Enter make all to generate benchmark_selfeval_ctl.exe; 3. Run benchmark_selfeval_ctl.exe to collect statistics on the total time, average time, average CPU cycles, and throughput for key generation, encapsulation, and decapsulation; 4. The program records static/baseline memory usage before the algorithm runs, records peak memory usage during repeated execution, and finally generates the ctl_kem_benchmark_raw.csv and ctl_kem_benchmark_summary.csv result files.
Expected Results	The program can run normally and output the algorithm performance test results, including the average CPU cycles, throughput, static/baseline memory usage, and peak memory usage for key generation, encapsulation, and decapsulation operations.

Actual Results	The program can run normally and outputs performance test results. The program collects statistics on the average CPU cycles, throughput, static/baseline memory usage, and peak memory usage of key generation, encapsulation, and decapsulation operations. The number of test runs is 100. Key generation: 1567500215 cycles/op average, 1.4 ops/s throughput; Encapsulation: 147730 cycles/op average, 16388 ops/s throughput; Decapsulation 33467357 cycles/op average, 72 ops/s throughput. Static/baseline working set memory average: 7531165 Bytes; Peak memory average: 7611596.80 Byte;
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3.3. Transmission and Storage Overhead Test

Test ID	OPT-7
Test Name	CTL-257-512 Optimized Implementation Transmission and Storage Overhead Test
Test Objective	Collect statistics on the data sizes involved in the key encapsulation process of the CTL-257-512 optimized implementation, including the public key, private key, ciphertext, and shared key.
Preconditions	1. The CTL optimized implementation program code runs normally; 2. The algorithm interfaces can return or confirm the lengths of each type of data object; 3. The test program can output transmission and storage overhead statistics.
Test Steps	1. Enter the CTL-257-512 optimized implementation program code directory; 2. Enter make all to generate benchmark_selfeval_ctl.exe; 3. Run benchmark_selfeval_ctl.exe; 4. Record the public key length, private key length, ciphertext length, and shared key length;
Expected Results	The lengths of all related data objects of the CTL-257-512 optimized implementation can be obtained, in Bytes.
Actual Results	The program can run normally and outputs the transmission and storage overhead test results. The program separately collects the public key length, private key length, ciphertext length, and shared key length. The number of test runs is 100. Public key length: 521 Bytes; Private key length: 2,953 Bytes; Ciphertext length: 473 Bytes; Shared secret value length: 16 Bytes.

Test ID	OPT-8
Test Name	CTL-769-1024 Optimized Implementation Transmission and Storage Overhead Test
Test Objective	Collect statistics on the data sizes involved in the key encapsulation process of the CTL-769-1024 optimized implementation, including the public key, private key, ciphertext, and shared key.
Preconditions	1. The CTL optimized implementation program code runs normally; 2. The algorithm interfaces can return or confirm the lengths of each type of data object; 3. The test program can output transmission and storage overhead statistics.
Test Steps	1. Enter the CTL-769-1024 optimized implementation program code directory; 2. Enter make all to generate benchmark_selfeval_ctl.exe; 3. Run benchmark_selfeval_ctl.exe; 4. Record the public key length, private key length, ciphertext length, and shared

	key length;
Expected Results	The lengths of all related data objects of the CTL-769-1024 optimized implementation can be obtained, in Bytes.
Actual Results	The program can run normally and outputs the transmission and storage overhead test results. The program separately collects the public key length, private key length, ciphertext length, and shared key length. The number of test runs is 100. Public key length: 1,230 Bytes; Private key length: 6,030 Bytes; Ciphertext length: 1,006 Bytes; Shared secret value length: 32 Bytes.

Test ID	OPT-9
Test Name	CTL-3329-2048 Optimized Implementation Transmission and Storage Overhead Test
Test Objective	Collect statistics on the data sizes involved in the key encapsulation process of the CTL-3329-2048 optimized implementation, including the public key, private key, ciphertext, and shared key.
Preconditions	1. The CTL optimized implementation program code runs normally; 2. The algorithm interfaces can return or confirm the lengths of each type of data object; 3. The test program can output transmission and storage overhead statistics.
Test Steps	1. Enter the CTL-3329-2048 reference implementation program code directory; 2. Enter make all to generate benchmark_selfeval_ctl.exe; 3. Run benchmark_selfeval_ctl.exe; 4. Record the public key length, private key length, ciphertext length, and shared key length;
Expected Results	The lengths of all related data objects of the CTL-3329-2048 optimized implementation can be obtained, in Bytes.
Actual Results	The program can run normally and outputs the transmission and storage overhead test results. The program separately collects the public key length, private key length, ciphertext length, and shared key length. The number of test runs is 100. Public key length: 3,009 Bytes; Private key length: 15,617 Bytes; Ciphertext length: 2,353 Bytes; Shared secret value length: 48 Bytes.